



Semester II 2019/2020

Subject : Bioinformatics I (SCSB2103)
Section : 01 – Dr Haslina Hashim
Topic : Lab 04 – BLAST

KOO XUAN (A23CS0300)

- 1) Perform a BLASTP search. In this problem we will explore the effect of a short protein query on the BLASTP parameters
 - i) Perform a BLASTP search at NCBI using the following query of just 12 amino acids.
By default, the parameters are adjusted for short queries.

(a) PNLHGLFGRKTG
 - ii) Inspect the output. What is the E-value cut off? What is the word size? What is the scoring matrix? How do these settings compare to the default parameters?

First , we enter PNLHGLFGRKTG for the query sequence :

The screenshot shows the NCBI BLASTP search interface. The 'Enter Query Sequence' field contains the text 'PNLHGLFGRKTG'. Below this, there are options to 'Choose File' or 'No file chosen'. The 'Choose Search Set' section has 'Standard databases (nr etc.)' selected. The 'Program Selection' section has 'blastp (protein-protein BLAST)' selected. The 'BLAST' button is visible at the bottom left of the form.

After click BLAST button, it will direct to here :

NIH National Library of Medicine
National Center for Biotechnology Information

BLAST® » blastp suite » RID-0E8U0PW0016

Home Recent Results Saved Strategies Help

[formatting options]

Format Request Status

Job Title: Protein Sequence

Your search parameters were adjusted to search for a short input sequence.

Request ID	0E8U0PW0016
Status	Searching
Submitted at	Tue Apr 22 01:04:00 2025
Current time	Tue Apr 22 01:04:16 2025
Time since submission	00:00:15

This page will be automatically updated in 7 seconds

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And automatically updated to here and we click search summary, and the results show:

BLAST® » blastp suite » results for RID-0E8U0PW0016

< Edit Search Save Search Search Summary

Search Parameters	
Program	blastp
Word size	2
Expect value	200000
Hitlist size	100
Gapcosts	9,1
Matrix	PAM30
Filter string	F
Genetic Code	1
Window Size	40
Threshold	11
Composition-based stats	0

Database	
Posted date	Apr 20, 2025 3:30 AM
Number of letters	345,751,102,228
Number of sequences	908,913,892
Entrez query	None

Karlin-Altschul statistics		
Lambda	0.356191	0.294
K	0.300975	0.11
H	1.89356	0.61
Alpha	0.1938	0.48
Alpha_v	0.161818	2.46392
Sigma		3.18615

Results Statistics	
--------------------	--

The large Expect value serves the function of allowing me to see results in which the result may be biologically meaningful, but the Expect value (related to a probability value) was not very small. E could not be small because, given a short query, there is

no opportunity for any alignment to achieve a high score (from $E = kmne(\exp)-1$ is a large score is associated with a small E value). The default Expect value is 10.

The matrix is PAM30 which is stringent relative to PAM250 or BLOSUM62 (the default matrix): mismatches are not well tolerated.

The word size is 2 although the default is 3.

2) Protein searches are usually more informative than DNA searches. Why is this so? Do a BLASTP search

- i) Do a BLASTP search using RBP4 (NP_006735), restricting the output to Arthropoda (insects).
- ii) Next, do a BLASTN search using the RBP4 nucleotide sequence (NM_006744). For this query, select only the nucleotides corresponding to the coding region of the DNA. (To do this visit the NCBI Nucleotide page, follow the link to the coding sequence [CDS], then choose the FASTA format.)
- iii) Which search is more informative? How many databases matches have an E value less than 1.0 in each search?

Set the settings for the BLASTP search:

The screenshot shows the NCBI Standard Protein BLAST search interface. The 'Enter Query Sequence' section has the accession number NP_006735 entered. The 'Choose Search Set' section has 'Database' set to 'Standard databases (nr etc.)', 'Organism' set to 'Arthropoda (taxid 6656)', and 'Exclude' options. The 'Program Selection' section has 'Algorithm' set to 'blastp (protein-protein BLAST)'. The 'BLAST' button is visible at the bottom.

There are about 100 matches having an E value less than 0.01 (while the problem asks about an E value <1 , it would have been better to ask about $E < 0.01$). Here are the first matches:

Sequences producing significant alignments									
Download Select columns Show 100 ?									
select all 100 sequences selected GenPept Graphics Distance tree of results Multiple alignment MSA Viewer									
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per Ident	Acc. Len	Accession
☒	apolipoprotein D [Anopheles darlingi]	Anopheles darlingi	60.8	60.8	89%	9e-09	28.02%	199	FIN60997.1
☒	outer membrane lipoprotein B1c-like [Anopheles albimanus]	Anopheles albimanus	60.5	60.5	86%	1e-08	28.98%	199	XP_035774284.1
☒	apolipoprotein D-like [Anopheles coustani]	Anopheles coustani	60.5	60.5	78%	1e-08	28.75%	200	XP_058121236.1
☒	unnamed protein product [Mediopia subpectinata]	Mediopia subpectinata	60.1	60.1	76%	1e-08	28.10%	203	CAD7635754.1
☒	unnamed protein product [Opieella nova]	Opieella nova	59.7	59.7	79%	2e-08	25.79%	200	CAD7641152.1
☒	hypothetical protein ZHAS_00021950 [Anopheles sinensis]	Anopheles sinensis	59.7	59.7	88%	2e-08	30.65%	200	KFB53445.1
☒	uncharacterized protein LOC126592005 [Anopheles darlingi]	Anopheles darlingi	60.8	159	95%	3e-08	28.02%	842	XP_049537156.1
☒	unnamed protein product [Mediopia subpectinata]	Mediopia subpectinata	60.1	60.1	64%	5e-08	29.46%	329	CAD76357136.1
☒	uncharacterized protein LOC126572729 [Anopheles aquasalis]	Anopheles aquasalis	60.5	161	94%	5e-08	28.16%	669	XP_050088230.1
☒	unnamed protein product [Opieella nova]	Opieella nova	58.5	58.5	79%	6e-08	25.79%	200	CAD7641151.1
☒	Apolipoprotein D [Araneus ventricosus]	Araneus ventricosus	57.8	57.8	69%	8e-08	28.26%	185	GBM98444.1
☒	apolipoprotein D-like [Calliphora vicina]	Calliphora vicina	57.8	57.8	76%	1e-07	30.92%	194	XP_065358042.1
☒	apolipoprotein D-like [Anopheles stephensi]	Anopheles stephensi	57.0	57.0	81%	2e-07	30.12%	202	XP_035912902.1
☒	insecticynin-A-like [Cydia splendana]	Cydia splendana	56.6	56.6	87%	3e-07	27.93%	210	XP_063628112.1
☒	Apolipoprotein D [Lucilia cuprina]	Lucilia cuprina	56.2	56.2	77%	3e-07	30.49%	194	KAI8130010.1
☒	insecticynin-A-like [Leguminivora glycinivorella]	Leguminivora glycinivorella	56.2	56.2	87%	4e-07	27.37%	210	XP_047994122.1
☒	apolipoprotein D-like [Athalia rosae]	Athalia rosae	55.8	55.8	80%	5e-07	26.38%	188	XP_012255906.2
☒	apolipoprotein D [Musca domestica]	Musca domestica	55.5	55.5	69%	7e-07	28.78%	192	XP_005180903.2
☒	apolipoprotein D isoform X3 [Bactrocera dorsalis]	Bactrocera dorsalis	55.1	55.1	57%	9e-07	30.58%	193	XP_049302741.1
☒	insecticynin-A-like [Cydia amplana]	Cydia amplana	55.5	55.5	87%	9e-07	27.37%	210	XP_063367265.1
☒	apolipoprotein D-like [Teleopsis dalmanni]	Teleopsis dalmanni	55.1	55.1	76%	9e-07	23.08%	191	XP_037954801.1
☒	apolipoprotein D-like [Lucilia sericata]	Lucilia sericata	54.7	54.7	56%	1e-06	34.19%	194	XP_037821945.1

Next find the DNA coding sequence. The BLASTP result has a link to the NCBI protein page; this page (http://www.ncbi.nlm.nih.gov/protein/NP_006735.2) has a link to the DNA (NM_006744.3):

Protein
Protein
Search
Advanced
Help

GenPept
Send to:
Change region shown
Customize view
Analyze this sequence
Run BLAST
Identify Conserved Domains
Highlight Sequence Features
Find in this Sequence
Show in Genome Data Viewer
Protein 3D Structure
Crystal Structure of Retinol-Binding Protein 4 (RBP4) in complex with non-retinoid
PDB: 6QBA
Source: Homo sapiens, Saccharolobus solfataricus
Method: X-ray Diffraction
Resolution: 1.8 Å
See all 23 structures...
Articles about the RBP4 gene
Retinol and retinol binding protein 4 levels and COVID-19: a Mendelian r [BMC Pulm Med. 2024]
Retinol binding protein 4 and type 2 diabetes: from insulin resistance to pancr [Endocrine. 2024]

retinol-binding protein 4 isoform a precursor [Homo sapiens]
NCBI Reference Sequence: NP_006735.2
Identical Proteins FASTA Graphics
Go to:
LOCUS NP_006735 201 aa linear PRI 19-AUG-2024
DEFINITION retinol-binding protein 4 isoform a precursor [Homo sapiens].
ACCESSION NP_006735
VERSION NP_006735.2
DBSOURCE REFSEQ: accession NM_006744.4
KEYWORDS RefSeq; MANE Select.
SOURCE Homo sapiens (human)
ORGANISM Homo sapiens
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.
1 (residues 1 to 201)
FAN, J. and HU, J.
Retinol binding protein 4 and type 2 diabetes: from insulin resistance to pancreatic beta-cell function
Endocrine 85 (3), 1020-1034 (2024)
38520616
GeneRIF: Retinol binding protein 4 and type 2 diabetes: from insulin resistance to pancreatic beta-cell function.
Review article
2 (residues 1 to 201)
WANG, H., ZHANG, Z., XIE, L., LU, K., ZHANG, S. and XING, S.
Retinol and retinol binding protein 4 levels and COVID-19: a Mendelian randomization study
BMC Pulm Med 24 (1), 206 (2024)
38521384
GeneRIF: Retinol and retinol binding protein 4 levels and COVID-19: a Mendelian randomization study.
Publication Status: Online-Only

This is the features, from the features find CDS:

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National Center for Biotechnology Information

Log in

Nucleotide

Nucleotide

Search

Advanced

Help

FASTA

Send to

Homo sapiens retinol binding protein 4 (RBP4), transcript variant 1, mRNA
NCBI Reference Sequence: NM_006744.4
[GenBank](#) [Graphics](#)
>NM_006744.4:84-689 Homo sapiens retinol binding protein 4 (RBP4), transcript variant 1, mRNA
ATGAAGTGGGTGGGCGCTCTTGTCTGTGGCGGCTGGGACGCGCGCGGAGCGGACTGCCGAG
TGAGCAGCTTCGAGTCAAGGAGAAGTTCGACAAGGCTGCTTCTCTGGGACCTGGTACGCCATGGCCAA
GAAGGACCCGAGGGCCTCTTCTGACAGAACATCGTCGCGAAGTTCCTCGTGACGAGACCGGCCAG
ATGAGCGCCACAGCCAGGGCCGAGTCCGCTTTTGAATACTGGGACGTGTGCGACAGACATGGTGGCA
CCTTCACAGACACGAGGACCTGCGCAAGTTCAGATGAAGTACTGGGGCGTAGCCTCTTCTCCAGAA
AGGAATGATGACCACTGGTCTGTGACACAGACTACGACACATGCTGTCAGTACTCTCTGCGGCTC
CTGAACCTGATGGCAGCTGTGTGACAGTACTCTCTGCTGTTTCCCGGACCCCAACGGCTGCGCC
CAGAGCGCAGAAATTTGTAGGACGCGGAGGAGCTGTGCTGCGCAGGAGTACAGGCTGATCGT
CCACACGGTTACTGCGATGCGACATCAGAAAGAAACCTTTTGTAG

Change region shown

☐ Whole sequence
☒ Selected region
from: 84 to: 689

Update View

Customize view

Analyze this sequence

Run BLAST

Pick Primers

Show in Genome Data Viewer

Articles about the RBP4 gene

Extracellular proximal interaction profiling by cell surface-targeted TurboID reveals [Sci Signal. 2024]
Retinol and retinol binding protein 4 levels and COVID-19: a Mendelian r [BMC Pulm Med. 2024]
Retinol binding protein 4 and type 2 diabetes: from insulin resistance to pancr [Endocrine. 2024]

See all...

Search

ENG US 10:48 PM 22/4/2025

Use this as a query in a BLASTN search. Note the sequence range is automatically specified to correspond to the CDS:

blastn

blastp

blastx

tblastn

tblastx

Standard Nucleotide BLAST

BLASTN programs search nucleotide databases using a nucleotide query. more...

Reset page

Bookmark

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) Clear

NM_006744.3

From 85 To 690

Or, upload file

Choose File

No file chosen

Job Title

Enter a descriptive title for your BLAST search

☐ Align two or more sequences [?](#)

Choose Search Set

Database
☒ Standard databases (nr etc.) ☐ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases

Optional

Nucleotide collection (nr/nt) [?](#)

☒ arthropods (taxid 6656) ☐ exclude [Add organism](#)
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Optional

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Optional

☐ Sequences from type material

Optional

Entrez Query [?](#)
Enter an Entrez query to limit search [?](#) [YouTube](#) [Create custom database](#)

Program Selection

Optimize for
☐ Highly similar sequences (megablast)
☐ More dissimilar sequences (discontinuous megablast)
☒ Somewhat similar sequences (blastn)
Choose a BLAST algorithm [?](#)

BLAST

Search database nt using Blastn (Optimize for somewhat similar sequences)
☐ Show results in a new window

25°C

Mostly cloudy

Search

ENG US 11:17 PM 22/4/2025

The result is as follows:

BLAST® » blastn suite » results for RID-0FCT0YGN013

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[Edit Search](#) [Save Search](#) [Search Summary](#) [How to read this report?](#) [BLAST Help Videos](#) [Back to Traditional Results Page](#)

Info Your search is limited to records that include: arthropods (taxid:6656)

Job Title [refNM_006744.3](#)

RID [0FCT0YGN013](#) Search expires on 04-23 23:17 pm [Download All](#)

Program BLASTN [Citation](#)

Database nt [See details](#)

Query ID [NM_006744.3](#)

Description Homo sapiens retinol binding protein 4 (RBP4), transcript \ ...

Molecule type rna

Query Length 606

Other reports [Distance tree of results](#) [MSA viewer](#)

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[Add organism](#)

Percent Identity to E value to Query Coverage to

[Filter](#) [Reset](#)

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

☐ select all 0 sequences selected

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☐ Eudonia angustea genome assembly, chromosome_18	Eudonia angustea	53.6	53.6	9%	0.034	81.48%	15881630	OZ208089.1
☐ PREDICTED: Frankiniella occidentalis uncharacterized LOC113206112 (LOC113206112)...	Frankiniella occidentalis	53.6	53.6	12%	0.034	77.78%	1587	XM_026422037.2
☐ Pamme aurita genome assembly, chromosome_Z	Pamme aurita	53.6	53.6	7%	0.034	86.36%	72712871	OX352289.1
☐ Lycaena phlaeas genome assembly, chromosome_8	Lycaena phlaeas	53.6	53.6	10%	0.034	79.66%	19027537	HG995171.1

GenBank Graphics Distance tree of results MSA Viewer

ENG US 11:20 PM 22/4/2025

There are no database matches having an Expect value <0.01 . The protein search is far more sensitive than the DNA search.

- 3) This problem introduces batch queries. It is possible to search many queries simultaneously, either using the web-based BLAST (as in this problem) or via locally installed BLAST+.

Mosses are plants of the phylum Bryophyta, including the non-seed plant *Physcomitrella patens* that had its genome sequenced (Rensing et al., 2008).

Do mosses have any globin proteins, and if so, which human globin(s) are they most closely related to?

- First obtain the accession numbers of all human globins. There are several approaches to doing this, including BLASTP using beta globin and neuroglobin as queries. Other approaches involve DELTA-BLAST (Chapter 5), or Pfam (Chapter 6). These accession numbers are provided in Web Document 4.7.
- Perform a BLASTP search using all accession numbers as queries, entering them into the query box. Restrict the output to RefSeq proteins of the mosses.
- Results for each query are shown (one at a time) via a pull-down menu. Currently there are significant, although distant matches of all human globins to moss proteins except for hemoglobin subunit mu. (See for example the match between human epsilon globin

and predicted moss protein XP_001786089.1 with an E value of 0.01. A BLASTP search with that moss protein confirms it is related to many annotated plant globins.) Notably, only one human protein (neuroglobin, NP_001030585.1) has very strong matches to moss proteins such as *P. patens* predicted protein XP_001764902.1 (E value $2e-10$, 27% identity across a span of 138 amino acid residues).

I have obtained a list of accession numbers of human RefSeq proteins that are globins by using HBB (NP_000509) in a DELTA-BLAST search, selecting “all” of the 15 significant results, clicking “download” and sending the results to a csv file, and parsing them to get this list:

The screenshot displays the NCBI BLAST suite interface for a Standard Protein BLAST search. The 'blastp' tab is selected. In the 'Enter Query Sequence' section, the accession number 'NP_000509' is entered. The 'Choose Search Set' section has 'Standard databases (nr etc.)' selected, with 'Reference proteins (refseq_protein)' chosen from the dropdown. The 'Organism' is set to 'Homo sapiens (taxid:9606)'. Under 'Program Selection', the 'Algorithm' is set to 'DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)'. The 'BLAST' button is visible at the bottom left of the search area. The Windows taskbar at the bottom shows the date as 24/4/2025 and time as 10:11 AM.

Program DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST) [Citation](#)

Database refseq_protein [See details](#)

Query ID [NP_000509.1](#)

Description hemoglobin subunit beta [Homo sapiens]

Molecule type amino acid

Query Length 147

Other reports [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

Type common name, binomial, taxid or group name
[+ Add organism](#)

Percent Identity to E value to Query Coverage to

PSI-BLAST incl. threshold 0.005 [Filter](#) [Reset](#)

Run PSI-Blast iteration 2

Number of sequences 500 [Run](#)

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments

15 sequences selected [GenPept](#) [Gra](#)

Sequences with E-value BETTER than threshold

☒ select all 15 sequences selected

Description	Scientific Name	Max Score	Identities	Positives	Score	Accession	Select for PSI blast	Used to build PSSM	Newly added
☒ hemoglobin subunit gamma-2 [Homo sapiens]	Homo sapiens	190	1	ASN.1	7	NP_000175.1	☒		
☒ hemoglobin subunit beta [Homo sapiens]	Homo sapiens	187	187	100%	26-61	100.00%	147	NP_000509.1	☒
☒ hemoglobin subunit gamma-1 [Homo sapiens]	Homo sapiens	187	187	100%	36-61	72.79%	147	NP_000550.2	☒
☒ hemoglobin subunit delta [Homo sapiens]	Homo sapiens	187	187	100%	46-61	93.20%	147	NP_000510.1	☒
☒ hemoglobin subunit epsilon [Homo sapiens]	Homo sapiens	182	182	100%	26-59	75.51%	147	NP_005321.1	☒
☒ hemoglobin subunit zeta [Homo sapiens]	Homo sapiens	151	151	97%	36-47	35.86%	142	NP_005323.1	☒

Download [FASTA \(complete sequence\)](#) [FASTA \(aligned sequences\)](#) [GenBank \(complete sequence\)](#) [Hit Table \(text\)](#) [Hit Table \(CSV\)](#) [Descriptions Table \(CSV\)](#) [XML](#)

PSI-BLAST iteration 1

Accession Select for PSI blast ☐ Used to build PSSM ☐ Newly added ☐

Feedback

And get the accession:

NP_000175.1
 NP_000509.1
 NP_000550.2
 NP_000510.1
 NP_005321.1
 NP_005323.1
 NP_000508.1
 NP_005322.1
 NP_001003938.1
 XP_005257062.1
 NP_599030.1
 NP_067080.1
 NP_001349775.1
 XP_016879605.1
 NP_001369741.1

Then performed a BLASTP search of the RefSeq database, restricting the organism to mosses.

BLAST® » blastp suite Home Recent Results Saved Strategies Help

blastn **blastp** blastx tblastn tblastx Standard Protein BLAST

BLASTP programs search protein databases using a protein query. more... Reset page Bookmark

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) Clear Query subrange

NP_067080.1
NP_001349775.1
XP_016879605.1
NP_001369741.1

From To

Or, upload file No file chosen ?

Job Title

☐ Align two or more sequences ?

Choose Search Set

Database ☒ Standard databases (nr etc.) ☐ Experimental databases

☒ Reference proteins (refseq_protein) ?

Organism exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown ?

Exclude ☐ Models (X/MXP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm ☒ blastp (protein-protein BLAST)
☐ PSI-BLAST (Position-Specific Iterated BLAST)
☐ PHI-BLAST (Pattern Hit Initiated BLAST)
☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm ?

Search database refseq_protein using Blastp (protein-protein BLAST)
☐ Show results in a new window

30°C Partly sunny Search ENG US 10:19 AM 24/4/2025

And edit the algorithm parameters:

Algorithm parameters Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign Restore default search parameters

General Parameters

Max target sequences ?
Select the maximum number of aligned sequences to display ?

Short queries ☒ Automatically adjust parameters for short input sequences ?

Expect threshold ?

Word size ?

Max matches in a query range ?

Scoring Parameters

Matrix ?

Gap Costs ?

Compositional adjustments ?

Filters and Masking

Filter ☐ Low complexity regions ?

Mask ☐ Mask for lookup table only ?
☐ Mask lower case letters ?

Search database refseq_protein using Blastp (protein-protein BLAST)
☐ Show results in a new window

10°C rty sunny Search ENG US 10:22 AM 24/4/2025

Here the output and it includes a pull-down menu allowing you to inspect the results from each protein query. The best results (shown below) are for human neuroglobin as a query; matches to moss proteins have convincing Expect values of 7.6, 14 and 18.

National Library of Medicine
National Center for Biotechnology Information

Log in

BLAST® » blastp suite » results for RID-0K84VW1B013

HomeRecent ResultsSaved StrategiesHelp

Edit Search
Save Search
Search Summary

How to read this report?
BLAST Help Videos
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Your search is limited to records that include: mosses (taxid:3208)

Job Title

NP_000175:hemoglobin subunit gamma-2 [Homo...

RID

0K84VW1B013

Search expires on 04-25 10:23 am

Download All

Results for

12.ref|NP_067080.1| neuroglobin [Homo sapiens](151aa)

Program

1.ref|NP_000175.1| hemoglobin subunit gamma-2 [Homo sapiens](147aa)

Database

2.ref|NP_000509.1| hemoglobin subunit beta [Homo sapiens](147aa)

Query ID

3.ref|NP_000550.2| hemoglobin subunit gamma-1 [Homo sapiens](147aa)

Description

4.ref|NP_000510.1| hemoglobin subunit delta [Homo sapiens](147aa)

Molecule type

5.ref|NP_005321.1| hemoglobin subunit epsilon [Homo sapiens](147aa)

Query Length

6.ref|NP_005323.1| hemoglobin subunit zeta [Homo sapiens](142aa)

Other reports

7.ref|NP_000508.1| hemoglobin subunit alpha [Homo sapiens](142aa)

Descriptions

8.ref|NP_005322.1| hemoglobin subunit theta-1 [Homo sapiens](142aa)

Sequences pr

9.ref|NP_001003938.1| hemoglobin subunit mu [Homo sapiens](141aa)

select all

10.ref|XP_005257062.1| cytoglobin isoform X1 [Homo sapiens](202aa)

Descriptions

11.ref|NP_599030.1| cytoglobin [Homo sapiens](190aa)

Descriptions

12.ref|NP_067080.1| neuroglobin [Homo sapiens](151aa)

Descriptions

13.ref|NP_001349775.1| myoglobin isoform 1 [Homo sapiens](154aa)

Descriptions

14.ref|XP_016879605.1| cytoglobin isoform X2 [Homo sapiens](137aa)

Descriptions

15.ref|NP_001369741.1| myoglobin isoform 2 [Homo sapiens](99aa)

Filter Results

Organism

only top 20 will appear

exclude

Type common name, binomial, taxid or group name

Add organism

Percent Identity

E value

Query Coverage

Filter

Reset

Download

Select columns

Show 100

GenPeptGraphicsDistance tree of resultsMultiple alignmentMSA Viewer

Scientific Name

Max Score

Total Score

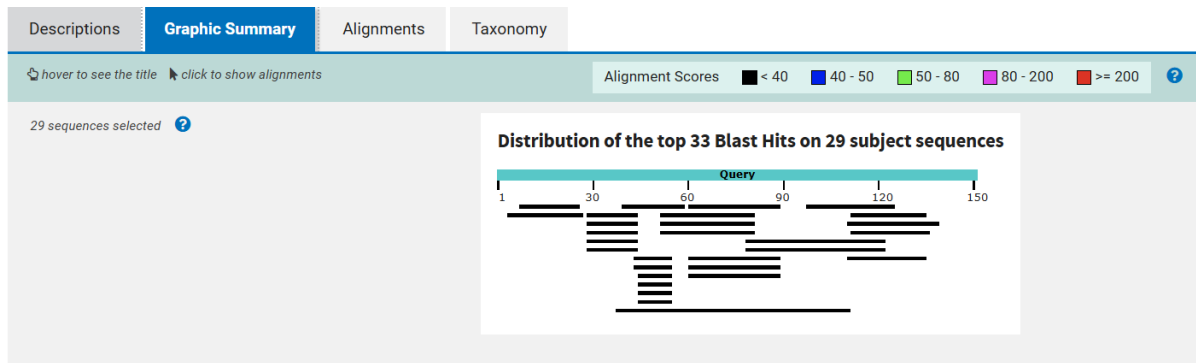
Query Cover

E value

Per Ident

Acc. Len

Accession



select all 29 sequences selected

GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ uncharacterized protein LOC112285949 [Physcomitrium patens]	Physcomitrium patens	26.9	26.9	13%	7.6	50.00%	270	XP_024383138.1
☒ DNA repair protein RAD5A-like [Physcomitrium patens]	Physcomitrium patens	26.6	26.6	19%	14	50.00%	1147	XP_024377727.1
☒ probable ectonucleotidase 53 [Physcomitrium patens]	Physcomitrium patens	25.8	25.8	16%	18	41.67%	418	XP_024372006.1
☒ paired amphipathic helix protein Sin3-like 3 isoform X2 [Physcomitrium patens]	Physcomitrium patens	25.8	25.8	11%	23	68.75%	1388	XP_024363995.1
☒ paired amphipathic helix protein Sin3-like 4 isoform X1 [Physcomitrium patens]	Physcomitrium patens	25.8	25.8	11%	23	68.75%	1390	XP_024363992.1
☒ paired amphipathic helix protein Sin3-like 3 [Physcomitrium patens]	Physcomitrium patens	24.3	24.3	11%	82	62.50%	1410	XP_024364100.1
☒ paired amphipathic helix protein Sin3-like 4 isoform X2 [Physcomitrium patens]	Physcomitrium patens	23.9	23.9	11%	86	62.50%	1391	XP_024382732.1
☒ paired amphipathic helix protein Sin3-like 4 isoform X1 [Physcomitrium patens]	Physcomitrium patens	23.9	23.9	11%	87	62.50%	1394	XP_024382731.1
☒ probable ectonucleotidase 53 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	19%	190	31.03%	371	XP_024376497.1
☒ pyruvate decarboxylase 2-like isoform X2 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	8%	201	50.00%	531	XP_024356345.1
☒ uncharacterized protein LOC112283610 isoform X1 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	20%	201	46.67%	827	XP_024378302.1
☒ pyruvate decarboxylase 2-like isoform X1 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	8%	203	50.00%	579	XP_024356342.1
☒ uncharacterized protein LOC112283610 isoform X3 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	20%	205	46.67%	826	XP_024378304.1
☒ uncharacterized protein LOC112283610 isoform X2 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	20%	205	46.67%	826	XP_024378303.1
☒ importin-4-like [Physcomitrium patens]	Physcomitrium patens	22.3	22.3	17%	309	36.00%	1049	XP_024374755.1
☒ zinc finger CCH domain-containing protein 14-like isoform X2 [Physcomitrium patens]	Physcomitrium patens	21.9	43.1	7%	335	54.55%	433	XP_024396649.1
☒ zinc finger CCH domain-containing protein 14-like isoform X1 [Physcomitrium patens]	Physcomitrium patens	21.9	43.1	7%	339	54.55%	438	XP_024396648.1
☒ zinc finger CCH domain-containing protein 14-like isoform X3 [Physcomitrium patens]	Physcomitrium patens	21.9	43.1	7%	341	54.55%	431	XP_024396650.1
☒ zinc finger CCH domain-containing protein 14-like isoform X4 [Physcomitrium patens]	Physcomitrium patens	21.9	43.1	7%	343	54.55%	388	XP_024396651.1
☒ kinesin-like protein KIN-14N [Physcomitrium patens]	Physcomitrium patens	21.6	21.6	29%	491	29.55%	795	XP_024381679.1
☒ fumarylacetoacetase-like [Physcomitrium patens]	Physcomitrium patens	21.6	21.6	13%	532	42.11%	425	XP_024372480.1
☒ mechanosensitive ion channel protein 2, chloroelastic-like isoform X3 [Physcomitrium patens]	Physcomitrium patens	21.6	21.6	19%	582	37.93%	546	XP_024360340.1
☒ kinesin-like protein KIN-14N [Physcomitrium patens]	Physcomitrium patens	21.2	21.2	29%	609	29.55%	797	XP_024389770.1
☒ mechanosensitive ion channel protein 2, chloroelastic-like isoform X1 [Physcomitrium patens]	Physcomitrium patens	21.2	21.2	19%	635	37.93%	582	XP_024360326.1
☒ mechanosensitive ion channel protein 2, chloroelastic-like isoform X2 [Physcomitrium patens]	Physcomitrium patens	21.2	21.2	19%	667	37.93%	558	XP_024360334.1
☒ uncharacterized protein LOC112288018 [Physcomitrium patens]	Physcomitrium patens	20.8	20.8	16%	926	33.33%	328	XP_024387532.1
☒ mechanosensitive ion channel protein 2, chloroelastic-like isoform X4 [Physcomitrium patens]	Physcomitrium patens	20.8	20.8	19%	944	37.93%	437	XP_024360348.1
☒ dnaJ homolog subfamily C GRV2-like isoform X1 [Physcomitrium patens]	Physcomitrium patens	20.8	20.8	49%	977	27.63%	2628	XP_024373332.1
☒ probable ectonucleotidase 53 [Physcomitrium patens]	Physcomitrium patens	20.8	20.8	17%	983	32.00%	363	XP_024385446.1

Feedback

ENG US 10:38 AM 24/4/2025

4) Is the pol protein of HIV-1 more closely related to the pol protein of HIV-2 or to the pol protein of simian immunodeficiency virus (SIV)? Use the BLASTP program to decide.

Hint: try the Entrez command “NOT hiv-1[organism]” to focus the search away from HIV-1 matches.

Go to website (<http://www.ncbi.nlm.nih.gov/protein/28872819?report=fasta>) and click on protein click on fasta to get this:

```
>NP_057849.4 Gag-Pol [Human immunodeficiency virus 1]
MGARASVLSGGELDRWEKIRLRPGGKKKYKCLKHIVWASRELERFAVNPGLLETSEGCRCQILGQLQPSLOT
GSEELRSLYNTVATLYCVHQRIEIKDTKEALDKIEEEQNKSKKKAQQAADTGHNSQVSQNYPIVQNIQG
QMVHQAISPRTLNAWVKVVEEKAFSPEVIMFSAALSEGATPQDLNLTMLNTVGGHQAAMQMLKETINEEAA
EWDRVHPVHAGPIAPGQMREPRGSDIAGTTSTLQEQIGWMTNNPPIPVGEIYKRWIILGLNKIVRMYSP
SILDIRQGPKEPFRDYVDRFYKTLRAEQASQEVKNWMTETLLVQANPDCKTILKALGPAATLEEMMTAC
QGVGGPGHKARVLAEAMSQVTNSATIMMQRGNFNRQKIVKCFNCGKEGHTARNCRAPRKKGCWKCGKEG
HQMKDCTERQANFLREDLAFLOQKAREFSSEQTRANSPTRRELQVWGRDNNSPSEAGADRQGTVSFNFPO
VTLWQRPLVTIKIGGQLKEALLDGTGADDTVLEEMSLPGRWPKPMIGGIGGFIVRQYDQILIEICGHKAI
GTVLVGPTPVNIIGRNLLTQIGCTLNFPISPIETVPVKLKPGMDGPKVKQWPLTEEKIKALVEICTEMEK
EGKISKIGPENPYNTPVFAIKKKDSTKWRKLVDFRELNKRTQDFWEVQLGIPHPAGLKKKKSVTVLVDVG
AYFSVPLDEDFRKYTAFTIPIINNETPGIRYQYNVLPQGWKGSPAIFQSSMTKILEPFRKQNPDIYIQQY
MDDLIVGSDLEIGQHRTKIEELRQHLLRWGLTTPDKKHQKEPFLWMGYELHPDKWTVQPIVLPEDKSWT
VNDIQKLVGKLNWASQIYPGIKVRQLCKLLRGTKALTEVIPLTEEALELAENREILKEPVHGVYDPSK
DLIAEIQKQGQGWYQIYQEPFKNLKTGKYARMRGHAHTNDVKQLTEAVQKITTESIWIWGKTPKFKLPI
QKETWETWWTYQATWIPWFEFVNTFPLVKLWYQLEKEPIVGAETFYVDGAANRETKLGKAGYVTNRGR
```

QKVVTLTDTTNQKTELQAIYLLALQDSGLEVNIVTDSQYALGIIQAQPDQSESELVNQIIIEQLIKKEKVVYL
 AWVPAHKGIGGNEQVDKLVSAGIRKVLFLDGDIDKAQDEHEKYHSNWRAMASDFNLPVVAKEIVASCDKC
 QLKGEAMHGQVDCSPGIWQLDCTHLEGGVILVAVHVASGYIEAEVIPAETGQETAYFLLKLAGRWPVKTI
 HTDNGSNFTGATVRAACWWAGIKQEFGIYPNPQSQGVVESMNKELKKIIGQVRDQAEHLKTAVQMAVFIH
 NFKRKGIGGYSAGERIVDIIATDIQTKELQKQITKIQNFRVYRDSRNPWLKGPAKLLWKGEAVVIQD
 NSDIKVVPRRKAIIIRDYGKQMAGDDCVASRQDED

Standard Protein BLAST

blastn **blastp** blastx tblastn tblastx

BLASTP programs search protein databases using a protein query. more...

Reset page Bookmark

Enter Query Sequence

Enter accession number(s), g(s), or FASTA sequence(s) [?](#) [Clear](#)

NP_057849.4

Query subrange

From To

Or, upload file [Choose File](#) No file chosen [?](#)

Job Title

NP_057849 Gag-Pol [Human immunodeficiency virus 1]

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database

☒ Standard databases (nr, etc.) ☐ Experimental databases

Reference proteins (refseq_protein) [?](#)

Organism

HIV-1 (taxid:11676) [?](#) [exclude](#) [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude

☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WPI) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

BLAST

Search database refseq_protein using Blastp (protein-protein BLAST)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with [?](#) sign

+ Algorithm parameters

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10:44 AM 24/4/2025

Here is the result:

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? Your search is limited to records that exclude: HIV-1 (taxid:11676)

Job Title NP_057849:Gag-Pol [Human immunodeficiency virus 1]

RID [0K9C6URT013](#) Search expires on 04-25 10:44 am [Download All](#)

Program BLASTP [Citation](#)

Database refseq_protein [See details](#)

Query ID [NP_057849.4](#)

Description Gag-Pol [Human immunodeficiency virus 1]

Molecule type amino acid

Query Length 1435

Other reports [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[Add organism](#)

Percent Identity E value Query Coverage

to to to

Filter Reset

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments

Download Select columns Show 100

☒ select all 100 sequences selected

GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ gag-pol (Simian immunodeficiency virus)	Simian immunod...	1687	1687	99%	0.0	57.69%	1472	NP_687035.1
☒ gag-pol fusion polyprotein (Human immunodeficiency virus 2)	Human immunod...	1675	1675	99%	0.0	56.33%	1550	NP_663784.1
☒ pol protein, partial (Simian immunodeficiency virus SIV-mnd 2)	Simian immunod...	1377	1377	67%	0.0	68.68%	1010	NP_758887.1
☒ gag polyprotein (Human immunodeficiency virus 2)	Human immunod...	523	523	30%	2e-167	58.68%	521	NP_056837.1
☒ gag protein (Simian immunodeficiency virus)	Simian immunod...	502	502	30%	8e-160	57.08%	513	NP_054369.1
☒ gag protein (Simian immunodeficiency virus SIV-mnd 2)	Simian immunod...	486	486	32%	1e-153	53.18%	517	NP_758886.1
☒ pol polyprotein, partial (Feline immunodeficiency virus)	Feline immunod...	489	489	58%	2e-148	47.30%	996	NP_040973.1
☒ gag-pol precursor (Jembrana disease virus)	Jembrana disea...	497	687	97%	3e-147	30.37%	1432	NP_042683.1
☒ pol polyprotein (Puma lentivirus 14)	Puma lentivirus...	429	675	64%	2e-125	37.98%	1086	YP_009507791.1
☒ pol polyprotein (Equine infectious anemia virus)	Equine infectio...	424	605	65%	2e-123	39.88%	1138	NP_056902.1
☒ pol protein (Canine arthritis encephalitis virus)	Canine arthritis...	418	533	64%	2e-121	39.30%	1105	NP_040939.1
☒ pol protein, partial (Ovine lentivirus)	Ovine lentivirus	402	552	65%	6e-116	38.24%	1086	YP_009268869.1
☒ pol polyprotein (Visna-maedi virus)	Visna-maedi virus	392	535	65%	5e-112	37.39%	1101	NP_040840.1

10:55 AM 24/4/2025

Both proteins are comparably related. The SIV protein and HIV 2 has the same higher coverage (99% versus 99%) and both have E values of 0.

5) You perform a BLAST search and a result has an E value of about 1×10^{-4} .

What does this E value mean? What are some parameters on which an E value depends?

An E value (Expect value) of 1×10^{-4} in a BLAST search means that the probability of obtaining an alignment with a similar or better score purely by chance is 1 in 10,000. This suggests that the match is statistically significant and unlikely to have occurred randomly. Therefore, the null hypothesis that the alignment occurred by chance can be rejected, supporting the idea that the sequences are homologous. The E value depends on several parameters, including the alignment score, the length of the query sequence, the size of the database searched, the substitution matrix used, and the gap penalties, all of which influence how alignment scores are calculated.